

Application of Derivatives: Formula Sheet

Class 12 Mathematics – Chapter 6 Complete Reference & Formulas

1 Rate of Change of Quantities

- **Rate of Change Formula:** If $y = f(x)$, then the instantaneous rate of change of y with respect to x at $x = x_0$ is:

$$\left. \frac{dy}{dx} \right|_{x=x_0} = f'(x_0)$$

- **Parametric Rate of Change (Chain Rule):** If $x = f(t)$ and $y = g(t)$, then:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \quad \left(\frac{dx}{dt} \neq 0 \right)$$

- **Economic Applications:**

- **Marginal Cost (MC):** Instantaneous rate of change of total cost $C(x)$ with respect to output x :

$$MC = \frac{dC}{dx}$$

- **Marginal Revenue (MR):** Instantaneous rate of change of total revenue $R(x)$ with respect to units sold x :

$$MR = \frac{dR}{dx}$$

2 Increasing and Decreasing Functions

- **Analytical Definitions on Interval I :**

- **Increasing:** $x_1 < x_2 \implies f(x_1) \leq f(x_2), \quad \forall x_1, x_2 \in I$
- **Strictly Increasing:** $x_1 < x_2 \implies f(x_1) < f(x_2), \quad \forall x_1, x_2 \in I$
- **Decreasing:** $x_1 < x_2 \implies f(x_1) \geq f(x_2), \quad \forall x_1, x_2 \in I$
- **Strictly Decreasing:** $x_1 < x_2 \implies f(x_1) > f(x_2), \quad \forall x_1, x_2 \in I$

- **Derivative Test (Theorem 1):** Let f be continuous on $[a, b]$ and differentiable on (a, b) :

$$\begin{aligned} f'(x) &\geq 0, \forall x \in (a, b) \implies f \text{ is increasing on } [a, b] \\ f'(x) &> 0, \forall x \in (a, b) \implies f \text{ is strictly increasing on } [a, b] \\ f'(x) &\leq 0, \forall x \in (a, b) \implies f \text{ is decreasing on } [a, b] \\ f'(x) &< 0, \forall x \in (a, b) \implies f \text{ is strictly decreasing on } [a, b] \\ f'(x) &= 0, \forall x \in (a, b) \implies f \text{ is a constant function on } [a, b] \end{aligned}$$

3 Critical Points & Maxima/Minima Tests

- **Critical Points:** A point $x = c$ in the domain of f where either $f'(c) = 0$ or f is not differentiable.
- **First Derivative Test (Theorem 3):** Let f be continuous at a critical point c :
 - **Local Maxima:** $f'(x)$ changes sign from **positive to negative** as x increases through c ($f'(x) > 0$ to the left, $f'(x) < 0$ to the right).
 - **Local Minima:** $f'(x)$ changes sign from **negative to positive** as x increases through c ($f'(x) < 0$ to the left, $f'(x) > 0$ to the right).
 - **Point of Inflection:** $f'(x)$ does not change sign as x passes through c (neither a local maximum nor a local minimum).

4 Second Derivative Test & Absolute Extrema

- **Second Derivative Test (Theorem 4):** Let f be twice differentiable at c such that $f'(c) = 0$:

$$f''(c) < 0 \implies x = c \text{ is a point of } \mathbf{\text{local maxima}} \text{ (} f(c) \text{ is local max)}$$

$$f''(c) > 0 \implies x = c \text{ is a point of } \mathbf{\text{local minima}} \text{ (} f(c) \text{ is local min)}$$

$$f''(c) = 0 \implies \text{Test fails (revert to First Derivative Test)}$$

- **Absolute (Global) Maxima and Minima on Closed Interval $[a, b]$:** *Working Rule:*

1. Find all critical points $c_i \in (a, b)$ where $f'(c_i) = 0$ or f' is undefined.
2. Calculate function values at all critical points and interval boundary endpoints: $\{f(a), f(b), f(c_1), \dots\}$
3. Identify the highest and lowest values:

$$\text{Absolute Maximum Value} = \max\{f(a), f(b), f(c_1), f(c_2), \dots\}$$

$$\text{Absolute Minimum Value} = \min\{f(a), f(b), f(c_1), f(c_2), \dots\}$$